

Assignment 3 — Bridge Pattern

Student: Anarov Daniyal · **Group:** SE-2530 · **Course:** ShP-2216 Software Design Patterns · **Topic:** A — Drawing

Repository: <https://github.com/Collector119/asik-aitu-sdp3> · **Base commit:** 8b597af · **Extension commit:** 04c9886 · **Submitted commit:** df90a36 (the only later commit adds this hash to the report)

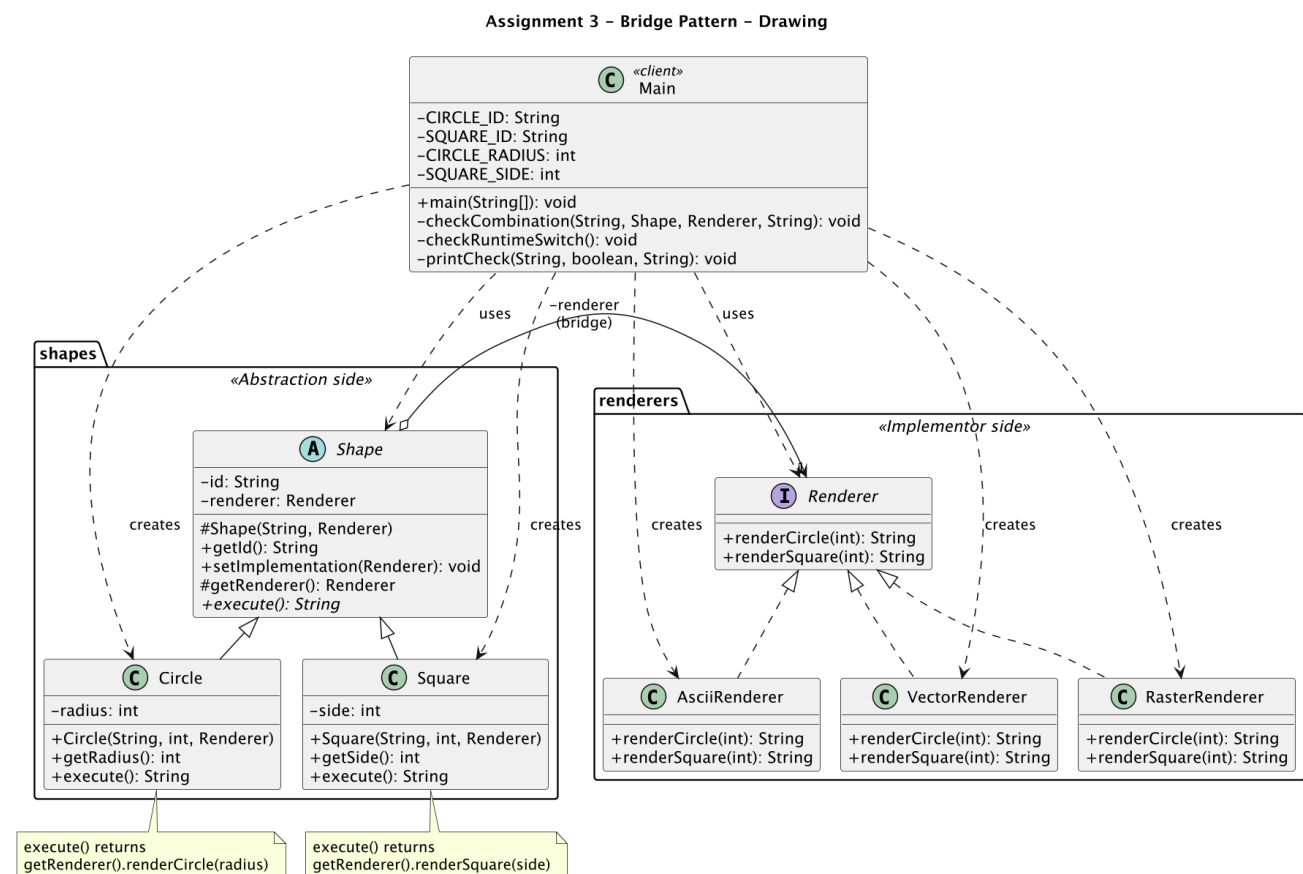
1. Two dimensions and why Bridge fits

My program draws shapes. It changes in two directions that do not depend on each other:

- **what** is drawn — the shape: Circle (radius 2) and Square (side 3);
- **how** it is drawn — the renderer: VectorRenderer, RasterRenderer and later AsciiRenderer.

If I solved this with inheritance only, I would need a class for every pair: VectorCircle, RasterCircle, VectorSquare, RasterSquare, and two more for ASCII. Every new shape or renderer multiplies the number of classes. Bridge splits the two directions into two hierarchies. The abstraction Shape stores a reference of the interface type Renderer, which it gets through its constructor, and passes the low-level drawing work to it. So 2 shapes and 3 renderers need 2 + 3 classes instead of 2×3 , and the renderer of an existing shape can be replaced at runtime with `setImplementation(...)`.

2. UML class diagram



3. Clean Code excerpts

Separation of responsibilities. A shape knows its own data, a renderer knows only how to draw. Circle does not build any text itself, it only chooses which low-level operation to call:

```
public String execute() {
    return getRenderer().renderCircle(radius);
}
```

Benefit: changes to the output format stay inside the renderers, and changes to a shape's data stay inside the shape.

Meaningful names. The names repeat the roles of the assignment, so a reader can find each role without comments: Shape, Renderer, setImplementation(Renderer renderer), checkRuntimeSwitch(), and constants such as

```
private static final int CIRCLE_RADIUS = 2;
private static final int SQUARE_SIDE = 3;
```

Benefit: the sample values are written once and have a name, instead of magic numbers in seven places.

Small focused methods. Each method in Main does one job: checkCombination runs one shape/renderer pair, checkRuntimeSwitch runs T5, and printCheck only counts and prints:

```
private static void printCheck(String checkId, boolean passed, String details) {
    totalChecks++;
    if (passed) {
        passedChecks++;
    }
    System.out.println(checkId + " " + (passed ? "PASS" : "FAIL") + " | " + details);
}
```

Benefit: every method can be read and explained on its own.

Avoidance of duplication. Six of the seven checks have the same workflow — execute, compare, print. It is written once in checkCombination, and the checks differ only in their data:

```
checkCombination("T1", new Circle(CIRCLE_ID, CIRCLE_RADIUS, vector), vector,
    "VECTOR path: circle radius=2");
```

Benefit: T6 and T7 were added with two calls and no new checking code. The PASS/FAIL label and the summary are computed in one place.

Encapsulation of state. The ID and the dimensions are private final, so they cannot change after construction. The bridge field is private; subclasses read it through getRenderer(), and only setImplementation can replace it:

```
private final String id;
private Renderer renderer;
```

Benefit: T5 can rely on the fact that switching the renderer cannot touch the ID or the radius.

4. Adding I3, trade-off and comparison with Adapter

How I3 was added. First I committed the working version with two renderers (8b597af). Then I added AsciiRenderer, which implements the same two methods of Renderer, and added T6 and T7 to Main (04c9886). extension.diff shows that in src/ only AsciiRenderer.java (new) and Main.java changed. Shape, Circle, Square, Renderer, VectorRenderer and RasterRenderer stayed unchanged, because shapes depend only on the Renderer interface and never on a concrete class.

Trade-off. Bridge makes adding a renderer cheap, but adding a shape is not free: the Renderer interface has one method per shape (renderCircle, renderSquare). A new Triangle would need a new renderTriangle method in the interface and in all three renderers. The design also has one more level of indirection than a single class would have.

Bridge vs Adapter. Both patterns use composition and an interface, but the intent is different. Adapter is applied *after* the classes exist: it wraps a class with an incompatible interface so that a client can use it. Bridge is planned *up front*: Shape and Renderer were designed together, so that both sides can grow independently. In my program nothing is incompatible — AsciiRenderer was written to implement

Renderer directly, so no adapter was needed. An adapter would be needed if I had to use an existing third-party drawing class with different method names.

References

1. Lecture 4, Bridge Pattern, ShP-2216 Software Design Patterns, Astana IT University, 2026.
2. E. Gamma, R. Helm, R. Johnson, J. Vlissides. *Design Patterns: Elements of Reusable Object-Oriented Software*. Addison-Wesley, 1994 — Bridge and Adapter.
3. A. Shvets. *Bridge*. Refactoring.Guru, <https://refactoring.guru/design-patterns/bridge>